



Underwater communication technologies: a review

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Accepted: 11 March 2025 / Published online: 19 April 2025
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Abstract

This review examines current underwater communication technologies, highlighting the challenges and innovations in applications spanning scientific research, exploration, environmental monitoring, and security. Emphasis is placed on the evolution and interplay of acoustic, optical, quantum, and hybrid communication methods, as well as their respective limitations and potential solutions in the complex underwater environment. The review explores advancements in Autonomous Underwater Vehicles (AUVs), with particular focus on the swarm configuration, which enables dynamic, interconnected networks for real-time data exchange and adaptive responses to environmental changes. Key areas of innovation include the use of new materials, advanced sensor networks, and machine learning algorithms to enhance communication efficiency, security, and resilience under varying underwater conditions. The integration of swarm AUVs and Internet of Things (IoT) concepts is proposed to further expand underwater operational capabilities, making underwater communication systems more reliable, secure, and versatile.

Keywords Underwater communication · Autonomous Underwater Vehicles (AUVs) · Swarm · Internet of Underwater Things (IoUT) · Biomimetic technology · Machine learning

1 Introduction

In the modern era, underwater communication has become a critical research and development field due to its essential applications in science, exploration, security, and environmental protection. Despite its importance, underwater communication presents unique challenges due to the physical characteristics of the aquatic environment. Traditional communication methods, such as radio frequencies, are limited in their penetration in water, while optical and acoustic communication face obstacles like absorption, scattering, and reflection [1]. The need for reliable and efficient underwater communication has led to the development of new technologies that address these challenges. Researchers are continually exploring innovative approaches to enhance communication capabilities in this demanding environment. This includes advancements in acoustic and ultrasonic signal

usage, as well as the exploration of optical signals, quantum communication, and underwater drones and Autonomous Underwater Vehicles (AUVs), with potential applications extending into the Internet of Things (IoT) framework for advanced underwater research and applications [2, 3].

The significance of secure underwater communication extends across multiple sectors [4]. In exploration, reliable communication methods are essential to conduct safe and efficient underwater missions. In environmental protection, the ability to monitor and collect data from underwater ecosystems offers valuable insights into marine biodiversity conservation. In defense, secure and effective underwater communication is critical for national border surveillance and protection [5, 6]. In addition, industrial applications, such as underwater mining and petroleum exploration, rely on reliable communication technologies to manage their operations effectively [7, 8]. Although existing technologies provide solutions to many problems, researchers continue to look for approaches to overcome current challenges and improve underwater communication capabilities. Innovations in underwater communication may include the development of new technological platforms, the application of advanced algorithms [9–11], and the exploration of alternative communication methods inspired by nature.

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Additionally, research aims to examine the potential of using new materials, sensor technologies [12], and quantum communication [13] to enhance the security, reliability, and efficiency of underwater communication networks.

Existing technologies, while providing solutions to many issues, also have limitations. For instance, acoustic communication, though capable of long-distance transmission, is prone to signal dispersion and interference in the noisy underwater environment [14]. Optical communication [15, 16], on the other hand, offers higher transmission rates but suffers from limited range due to light absorption and refraction in water. Recognizing these limitations, bio-inspired approaches may offer a promising perspective for developing new underwater communication technologies [17–19]. Research in biomimicry seeks to replicate natural processes and phenomena observed in the underwater world to create more efficient and effective technologies. Examples include mimicking the mechanisms of communication of sea mammals through acoustic signals or using bioluminescence to generate optical signals in the dark depths of the ocean. The goal is to develop a system that can adapt and function effectively in the diverse and often challenging conditions of the underwater environment, addressing issues like variable water conductivity, obstacles, and the physical limitations of signal transmission underwater. Innovation in underwater communication is not limited to technology alone, but extends to its applications. The development of new communication methods could bring significant improvements in underwater research activities, marine resource monitoring and management, security and defense, as well as underwater exploration and exploitation. By focusing on a wide range of technologies, from traditional methods to recent innovations, the aim is to explore how new technologies can provide enhanced communication capabilities in this complex and challenging environment. Through technological advancements and ongoing research, the objective is to develop new methods that overcome existing barriers, offering reliable, efficient, and secure communication below the sea.

The following chapters aim to provide the reader with a comprehensive overview of current technologies, challenges, and innovations in underwater communication, offering a view of the possibilities that can open new horizons for the future of underwater research and applied activities. By integrating technology and innovation, we aim to enhance humanity's ability to explore, monitor, and exploit underwater environments in ways that were inconceivable just a few decades ago. Continued progress in underwater communication promises to reveal the secrets of the deep oceans and push the boundaries of human knowledge and understanding [20].

1.1 Motivation, organization, and contribution

Underwater communication technologies are crucial for various applications, including environmental monitoring, underwater exploration, and defense operations. However, the challenging underwater environment imposes significant limitations on existing communication methods, such as signal attenuation, interference, and energy constraints. While several reviews have addressed individual communication methods, there is a lack of a comprehensive study that integrates both existing technologies and future innovations in a unified framework. This paper aims to bridge this gap by providing a holistic review of current underwater communication technologies while also exploring emerging trends and research directions.

This paper is organized as follows: Sect. 2 provides a detailed overview of existing underwater communication technologies, including acoustic, optical, and radio frequency-based systems. Section 3 explores innovative proposals and cutting-edge technologies, such as AI-driven adaptive communication, biomimetic approaches, and quantum communication. Section 4 outlines the major research challenges and open issues that need further investigation. Section 5 presents a discussion on the implications of the aforementioned advancements.

The primary contributions of this review are as follows:

- The review provides a comprehensive comparison of existing underwater communication technologies, highlighting their respective advantages and trade-offs.
- The introduction of novel research directions, including AI-driven dynamic communication systems, bio-inspired communication models, and hybrid communication approaches.
- It integrates recent advancements in machine learning and artificial intelligence, emphasizing their role in optimizing underwater communication networks.
- A dedicated section on research challenges and open issues, providing insights into future research opportunities in underwater communication.

By addressing both the state-of-the-art technologies and the challenges that remain unresolved, this review aims to offer valuable insights for researchers and practitioners working in the field of underwater communication.

2 Existing technologies

Effective underwater communication is essential for numerous applications, including ocean exploration, environmental

monitoring, military operations, and underwater sensor networks. However, the unique properties of the underwater environment, such as high signal attenuation, scattering, and absorption, impose significant challenges on data transmission. To address these limitations, various communication technologies have been developed, each offering distinct advantages and constraints depending on operational requirements and environmental conditions.

This section provides a structured overview of the primary underwater communication technologies, encompassing well-established methods such as acoustic and optical communication, as well as emerging approaches such as quantum communication and passive detection systems. While acoustic communication remains the dominant technology for long-range transmissions, it suffers from low data rates and susceptibility to multipath effects. Optical communication, on the other hand, enables high-speed data transmission but is highly sensitive to water turbidity, limiting its range. Radio frequency and electromagnetic communication have potential applications in shallow-water environments, though their effectiveness is greatly reduced in deep-sea conditions. Additionally, novel techniques like quantum and neutrino-based communication offer promising future alternatives, yet their implementation remains in early research stages.

A comparative summary of these technologies is provided in Table 1, outlining their key characteristics, advantages, limitations, and typical applications. This comparison highlights the trade-offs between different methods, offering a clearer understanding of their respective roles and the scenarios in which they are most effectively deployed. The insights gained from this analysis serve as a foundation for evaluating current solutions and identifying areas for future advancements in underwater communication.

The following is an overview of the main technologies:

2.1 Acoustic communication

Acoustic communication lies at the core of underwater communications, serving a wide range of applications, from navigation and exploration to surveillance and research of underwater environments. Using sound waves, this method enables data transmission through water, making it ideal for overcoming the limitations posed by water to other forms of communication, such as electromagnetic waves.

However, despite its effectiveness over long distances, this method faces specific challenges, such as noise interference and signal reflection from the seabed or other underwater objects [98]. Research in acoustic communication aims to optimize performance through the development of advanced coding and transmission techniques, as well as the application of strategies to reduce noise and enhance signal perception in the aquatic environment.

An interesting example of acoustic communication application is the use of underwater acoustic buoys for monitoring marine mammals [98, 99] or tsunami prediction [100, 101]. These buoys are equipped with acoustic sensors that can record underwater acoustic signals, providing valuable data for the scientific community. Analyzing data from these buoys can help understand marine mammal migration patterns, monitor ocean health, and even provide early warnings for natural disasters like tsunamis.

Beyond scientific research [102] and environmental monitoring [103], acoustic communication is also crucial for supporting underwater operations, such as defense surveillance [104] and underwater mining [105]. The ability of underwater vehicles and equipment to communicate over long distances without surface support opens new possibilities for conducting complex missions with greater autonomy.

This method is prevalent due to its ability to penetrate water more effectively compared to electromagnetic or optical waves [106]. However, despite its advantages, acoustic communication faces challenges like low data rates, limited bandwidth, and high latency, while still offering the benefit of long transmission range [107].

A novel approach to addressing these limitations is the use of Orthogonal Frequency Division Multiplexing (OFDM) combined with Subcarrier Power Modulation (SPM) [108]. This approach aims to double the data rate in acoustic communication without increasing the bandwidth, while also reducing information loss during transmission. Combining OFDM with Subcarrier Power Modulation (OFDM-SPM) is an innovative concept that modulates the power of each subcarrier to encode additional information, thus doubling the data rate without a corresponding increase in bandwidth. This technique not only increases the efficiency of available spectrum usage but also enables the transmission of larger data volumes with lower power consumption [109].

In addition to OFDM-SPM, the research community is exploring other approaches to enhance acoustic communication. Designing advanced equalizers on the receiver side plays a crucial role in accurately retrieving the original signal, addressing the challenge of multipath propagation and signal loss due to water conductivity. Improving acoustic communication technologies remains an active research area, with the development of new techniques and algorithms aimed at further reducing noise influence and increasing transmission efficiency.

2.2 Optical communication—underwater optical networks (UWON)

Optical communication in underwater environments represents an advanced form of communication that leverages the power of light to transmit information between two or more points. Using lasers or other light sources, optical communi-

Table 1 Comparative table of underwater communication technologies

Communication type	Usage	Advantages	Limitations	Technical characteristics
Acoustic communication [9, 21]	Underwater networks, long-range communication	Long range	Low data rate	Speed: 5–10 kbps, Range: up to 20 km, Frequency: 10 Hz–100 kHz
Underwater Wireless Optical Communication (UWOC) [16, 22–24]	High-speed underwater networks	High speed, low power consumption	Short range, affected by turbidity	Speed: up to 1 Gbps, Range: 10–100 m, Frequency: 400–800 THz (visible light)
Radio frequency (RF) communication [25–35]	Specialized underwater applications	Medium range	High absorption in deep water	Speed: up to 100 Mbps, Range: 1–10 m, Frequency: 3–30 MHz
Quantum Communication [36–43]	Secure data transmission	Excellent security	Implementation complexity	Speed: 1–100 Mbps, Range: < 10 km, Frequency: Photon-based
Passive Communication Systems [21, 44–49]	Detection without active signal transmission	Low power consumption	Limited data rate	Speed: 1–5 kbps, Range: up to 5 km, Frequency: Depends on natural radiation
Neutrino-Based Communication [50–62]	Deep-sea and long-distance communication	Virtually unlimited range	High cost and complexity	Speed: Very slow (bps), Range: > 1000 km, Frequency: Extremely low
Wired Communication [63–71]	Stable data transmission	Very high speed	Limited mobility	Speed: > 10 Gbps, Range: Unlimited, Frequency: Depends on cable properties
Multiple-Input Multiple-Output (MIMO) Systems [72–76]	Capacity enhancement in underwater networks	Increased efficiency	Requires advanced systems	Speed: 10–100 Mbps, Range: 100–500 m, Frequency: Hz–GHz
Surface Acoustic Wave (SAW) [77–80]	Specialized low-power applications	Very low power consumption	Limited distance	Speed: 1–10 kbps, Range: up to 10 m, Frequency: 10 MHz–1 GHz
Adaptive Communication [81–84]	Adapting to changing environments	High efficiency	Technical complexity	Speed: 5–50 Mbps, Range: Variable (1 m–10 km), Frequency: Variable
Magnetic Induction [85–89]	Short-range applications	Low dependence on transmission medium	Limited range	Speed: 1–10 kbps, Range: 1–5 m, Frequency: Low (kHz)
Translational Acoustic-RF (TARF) [90–92]	Conversion of acoustic to RF communication	Combines advantages of acoustic and RF	Technical complexity	Speed: 5–50 Mbps, Range: 10 m–1 km, Frequency: Hz–GHz
Photoacoustic/Thermoacoustic Communication [93–97]	Underwater detection and applications	Low energy consumption, accuracy	Short range	Speed: 1–5 kbps, Range: 1–10 m, Frequency: Depends on source

Table presents a comparative analysis of various underwater communication technologies, emphasizing their respective advantages, limitations, and primary technical characteristics. Each method offers unique benefits depending on the operational needs, ranging from long-distance acoustic communication to high-speed optical and quantum-based approaches. Understanding the strengths and constraints of these technologies is crucial for optimizing their implementation in underwater networks

cation offers the advantage of high data transmission speeds, making it ideal for applications that require fast exchange of large information volumes, such as underwater video communication or high-definition image transmission [15, 16, 23, 110, 111].

An example of optical communication technology is the Underwater Optical Wireless Network (UOWN), which uses lasers to establish a communication network between

underwater devices [24, 112]. This technology can deliver unmatched transmission speeds compared to acoustic methods, but it faces challenges such as light absorption and scattering in water, limiting its effectiveness over short distances and often requiring direct line-of-sight between devices [113].

To overcome these limitations, recent research has focused on understanding and mitigating the effects of underwa-

ter optical signal degradation. One significant challenge is the impact of polydispersity in underwater particles, which affects the polarization and transmission of light waves. Recent studies have modeled these interactions and proposed techniques to improve signal propagation efficiency by compensating for scattering effects and optimizing light paths through complex underwater environments [114]. These advancements are particularly relevant for deep-sea exploration, where maintaining stable and high-speed optical communication remains a fundamental challenge.

In addition to optimizing light transmission properties, researchers are also developing hybrid communication systems that combine optical and acoustic communication to benefit from the advantages of both. Moreover, the introduction of adaptive beamforming, advanced modulation techniques, and novel photonic materials has the potential to significantly enhance the reach and robustness of optical underwater networks [20, 115, 116].

2.3 Radio frequencies (RF) and electromagnetic communication

Radio frequencies (RF) [25] and electromagnetic communication [26] present an intriguing area of research in underwater communication. Although electromagnetic waves do not propagate through water as easily as they do through air due to high absorption and energy loss, using low frequencies can provide a viable solution for applications over very short distances or for data transmission from the seabed to the surface. This opens new pathways for underwater communication, especially in scenarios where other forms, such as acoustic or optical, are not feasible or effective.

Currently, electromagnetic communication is the most widely used method for communication between military submarines and ground command centers and plays a significant role in national defense and security. Electromagnetic waves propagate relatively smoothly in air, water, and across air-water boundaries and have low Doppler shift, making them the first choice for submarine communication.

Nevertheless, electromagnetic waves face certain limitations. Since seawater is highly conductive, electromagnetic waves at 800 MHz have an attenuation of 825 dB/m in seawater [27, 28]. In contrast, the attenuation of acoustic waves in seawater is only 0.035 dB/m [29], a considerable difference. As a result, electromagnetic wave frequency must be very low to ensure sufficient communication distance. Communication with submarines at very high (VHF) or ultra-high (UHF) frequencies is nearly impossible [30]. Extremely Low Frequencies (ELF) and Very Low Frequencies (VLF) are mainly used in electromagnetic communication with submarines. However, low frequency results in low data rates, the need for extremely powerful base stations, and large antennas. For example, communication with submarines at a distance of

200 ms can achieve only 50 to 100 bps [31]. Many global military forces have built massive transmission stations for submarine electromagnetic communication, which consume significant energy but support only basic functions due to data limitations.

A technology example being explored for RF use in underwater environments is specialized underwater antennas that operate at extremely low frequencies, thus enabling communication over short distances underwater. These antennas can be used to transmit critical data from underwater monitoring systems or Autonomous Underwater Vehicles (AUVs) to the surface [32, 33].

In addition to conventional RF technologies, research also focuses on developing innovative solutions in electromagnetic communication for underwater use. Exploring electromagnetic waves at low frequencies, which offer better water penetration, is an exciting prospect. This may include the use of specific modulations and techniques that allow communication over longer distances or through denser aquatic environments [34, 35].

The development of such technologies could enable, for example, the use of underwater sensors and other devices that require less frequent data exchange or signal transmission in situations where other communication forms are not feasible.

2.4 Quantum communication

Quantum communication, one of the most advanced and secure methods of communication, is based on the principles of quantum mechanics and is rapidly evolving, offering unprecedented possibilities in the field of underwater communication. Using quantum bits (qubits), quantum communication enables information transmission in a manner that surpasses the limits of classical communication, making it ideal for secure data exchange in underwater environments, especially for military and other critical purposes [13].

Through significant achievements, such as quantum entanglement distribution, free-space quantum cryptography (over distances exceeding 100 kms), and successful implementation of underwater quantum communication experiments, the potential of quantum communication for even greater capabilities in the future is evident [36–38]. These advancements highlight quantum communication's ability to offer absolute security, ensuring that quantum communications cannot be intercepted or breached by traditional means. Quantum communication, therefore, provides a reliable solution for ensuring absolute data confidentiality and integrity, making it ideal for applications where security is the highest priority [39–43].

Enhancing quantum repeater technology and creating a quantum communication network could enable even longer communication distances and broader coverage, facilitating secure and efficient underwater communications on a global

scale. The future development and improvement of quantum communication technology are expected to lead to even more advanced applications, with the potential to extend communication distances and further enhance security [117, 118].

2.5 Passive communication systems

Passive underwater communication systems represent an important category of communication technologies, enabling the utilization of natural or artificial sound signals generated in the aquatic environment [44]. This approach allows for the collection and analysis of information without the need for active signal transmission, thus providing an advantage in terms of security and stealth in communication [21]. For instance, the detection of sound signals from marine mammals or the mechanical operation of other underwater vehicles can be used for location estimation or navigation purposes. The development of passive communication systems also aims to improve detection and analysis technologies for acoustic signals [45]. Through the use of advanced machine learning algorithms, it is possible to interpret data more accurately, enabling more efficient communication and a better understanding of the underwater environment [46].

Additionally, the development of these systems can help improve navigation systems [47, 48] for submarines and other Autonomous Underwater Vehicles (AUVs) [49], providing more precise and reliable orientation information underwater.

2.6 Neutrino detection-based communication

One of the major challenges in submarine communication arises from the complex marine environment. The communication methods previously mentioned encounter difficulties in creating and adapting transmission channels. However, due to the unique properties of neutrinos, it appears feasible to use them as a technology for underwater communication, without needing special accommodations for the transmission channel's specific characteristics [50, 51].

A neutrino is a type of lepton, with no electric charge, very low mass (less than a millionth of an electron's mass), and nearly reaches the speed of light. Its interaction with other matter is very weak, allowing it to easily penetrate human bodies, walls, mountains, and even entire planets, making it difficult to detect. With exceptional penetrative ability, the neutrino renders channel characteristics irrelevant for underwater communication, as submarines can receive neutrino signals anywhere in the ocean, significantly reducing the risk of interception.

Originally proposed for interstellar communication, neutrinos have also been considered for underwater applications. Experiments like K2K [52], NEMO [53, 54], MINOS [55], and ANTARES [56, 57] have become well-known for neu-

trino study. Using a high-energy proton accelerator, protons can be accelerated to generate high-energy beams of hundreds of billions of electron volts [58]. These beams are then used to bombard a target, producing unstable particles, which decay into neutrinos and other particles. The neutrinos then pass through a thick layer of steel that blocks charged particles, resulting in an isolated neutrino beam.

Cherenkov light detection can serve as a method for neutrino signal reception in aquatic environments [59]. When a high-speed neutrino collides with a proton in water, the proton absorbs energy from the neutrino and moves at very high speed. If the proton's speed exceeds the speed of light in water, Cherenkov light is produced and detected by light sensors.

For submarines, the simplest approach to detecting a neutrino signal is to convert the submarine itself into a detector: a modern submarine has a large surface area, thus providing an effective detection area. Thin detection modules can be used to cover most of the submarine's exterior, enabling this setup.

Neutrino communication has favorable characteristics, such as long-range, low attenuation, and high stability. However, neutrino transmission requires a large amount of energy to ensure the number of transmitted neutrinos. Moreover, the neutrino detector can be bulky, with a radius of more than 100 ms, making it impractical for transport by submarines. Additionally, Cherenkov light is weak, mainly due to the low interaction frequency of neutrinos and the nature of secondary particles produced. As a result, the neutrino detector can only function at ocean depths to avoid interference from ambient light.

Despite the challenges associated with producing and detecting neutrinos, such as high energy consumption and the large size of detectors, this technology has shown significant future potential. While implementing such a communication system faces substantial technical challenges, its potential applications are extensive, from environmental monitoring [60] and scientific research [61] to secure military communication [62].

2.7 Data transmission via cable technologies

Cable data transmission technologies represent a critical infrastructure for underwater communication and connectivity. Using fiber optics and other cable types, these systems provide high speed and reliability in data transmission, capable of supporting the requirements of complex and critical applications [63, 64]. Cable technology is indispensable in scenarios that require the transfer of large data volumes with absolute security and minimal latency, such as communication between underwater research stations, oil rigs, and other permanent seabed infrastructures [65].

One of the most interesting applications of this technology is the creation of underwater optical fiber networks connecting continents and islands, ensuring global communication and internet access [66]. These networks are vital for the digital economy and global communication, providing the backbone for seamless data transmission worldwide.

Fiber optics, due to its high speed and exceptional durability in underwater conditions, enables long-distance data transmission with minimal loss. This makes it ideal for deploying complex and extensive underwater networks capable of supporting applications such as high-definition video transmission, telemetry, and data management from underwater research activities [67].

Furthermore, fiber optics is resistant to environmental challenges like electromagnetic interference and saltwater corrosion, which usually affect other communication forms. This durability ensures long-term and reliable operation in underwater environments, making it ideal for permanent infrastructures like oil rigs [68] and research centers [69].

With the expansion of underwater fiber optic networks, the potential for further innovations in technology and applications continues to grow. Future applications may include more advanced transmission technologies that offer even higher speeds and security [70], as well as the potential to integrate with new forms of underwater robotic systems that could improve ocean research and surveillance [71].

2.8 Multiple input multiple output (MIMO) communication systems

The application of MIMO (Multiple Input Multiple Output) technology in underwater environments presents unique challenges [72]. Acoustic conditions and the presence of various obstacles, such as water and the seabed, impact transmission efficiency. However, by using multiple antennas for signal transmission and reception, MIMO systems can harness multipath propagation in the aquatic medium, improving communication efficiency and reliability. This enables the transmission of larger data volumes with lower error rates, which is crucial for applications such as underwater robotics, communication among Autonomous Underwater Vehicles (AUVs), and data exchange between underwater research stations [73].

A specific example of MIMO technology application in underwater environments is enhancing the accuracy and reliability of acoustic positioning systems [74]. By using multiple antennas, MIMO systems can process complex signals from different sources and more accurately determine the position and direction of underwater vehicles, improving mission safety and success.

Beyond basic applications, MIMO technology can also enhance communication in conditions where traditional systems fail due to high interference or long distances [75].

The ability of MIMO systems to increase data density and spectrum efficiency is particularly valuable in environments where interference and signal loss are common issues [76].

2.9 Surface acoustic wave (SAW) technology

Surface Acoustic Wave (SAW) technology represents an advanced approach in the development of underwater sensors and communication systems [77]. It utilizes acoustic waves that propagate along the surface of a solid material to transmit information or detect environmental parameters [78]. These sensors can operate underwater, providing accurate measurements for temperature, pressure, flow, and chemical composition, making them particularly valuable in applications such as ecological monitoring, seabed research, and infrastructure surveillance.

SAW systems are known for their high reliability and ease of integration into more complex systems [79]. They offer significant advantages over traditional technologies, such as reduced power consumption and enhanced durability in underwater conditions [80]. The flexibility of SAW technology allows for sensor adaptation to specific needs and applications, from marine life monitoring to underwater threat detection.

2.10 Adaptive communication systems

Adaptive underwater communication systems represent an innovative approach to underwater communication, offering the flexibility to adjust technological parameters according to the specific needs of each application [81]. This adaptability enables efficient power management and the selection of the most suitable transmission method depending on the underwater environment and its challenges [82]. For example, in situations where visibility is reduced due to turbidity, an adaptive communication system can automatically switch the transmission method from optical to acoustic, ensuring continuity and reliability of communication [83].

A key aspect of adaptive communication is the ability to estimate and optimize channel conditions in real time. Intelligent reflecting surfaces (IRS) and advanced channel estimation techniques have been explored to enhance the efficiency of underwater adaptive communication. Recent research has demonstrated that IRS-aided communication systems can dynamically adjust transmission properties to mitigate signal degradation and improve link reliability, particularly in challenging environments with multipath effects and high attenuation [119]. Such advancements are crucial in underwater networks where environmental variations significantly impact signal propagation.

Moreover, the development of adaptive communication systems aims to create more flexible and self-optimizing network infrastructures. By incorporating automated detection

and adaptation mechanisms, these systems can dynamically manage changes in the underwater environment, enhancing communication efficiency under varying conditions and use scenarios [84]. Future research in this area focuses on integrating machine learning models to further refine adaptation processes, enabling underwater networks to predict environmental changes and proactively adjust communication parameters for optimal performance.

2.11 Magnetic communication via magnetic fields

Magnetic communication is a relatively new technology that relies on the principle of electromagnetic induction to transfer information through the creation and alteration of magnetic fields [85]. This communication method offers significant advantages, such as low latency and the simplicity of antenna construction, making it particularly useful for applications requiring rapid data exchange in underwater environments [86–88].

In magnetic communication, the transmission process involves creating a magnetic field by an emitting coil in water, with its frequency, amplitude, or phase modulated to carry information. Reception is performed using a receiving coil also placed in the water, which captures the magnetic field signal. After amplification, filtering, and decomposition, the received magnetic signal is converted into an electrical signal that can be decoded [89].

Although magnetic communication faces challenges, such as high energy consumption and the size of detectors, this technology has the potential to open new avenues for communication in challenging environments. The unique ability of magnetic signals to penetrate water with minimal energy loss provides a viable solution to underwater communication problems, especially in situations where other communication methods, such as acoustic and optical, are ineffective.

2.12 Translational acoustic-RF communication (TARF)

Translational Acoustic-RF Communication (TARF) is an innovative communication method that combines two mediums for communication, enabling data transmission from underwater vessels to surface-level receivers. This method opens new prospects for bidirectional communication, although it currently operates primarily as a one-way communication channel [90].

The operating principle of TARF communication is based on the creation of minute vibrations on the water surface from underwater acoustic signals. An airborne radar above the water surface measures these micro-disturbances and decodes the information. This process presents significant challenges, such as detecting extremely small disturbances

and managing the effects of the natural variability of the water surface.

In summary, TARF communication still faces significant challenges and limitations. Firstly, this method, based on theoretical data, only supports communication between the submarine and a surface receiver, allowing data transmission from the submarine but not reception. This is a considerable limitation because the submarine must be able to receive external instructions and know its location. Secondly, the ocean environment presents far greater challenges compared to a simple water tank or pool, with waves reaching heights of 20 to 30 ms. For stable system operation under such conditions, active surface monitoring and improvement of the communication protocol are required. Thirdly, the system's alignment requires advanced alignment, as the receiver must locate the position of the underwater transmitter by scanning the water surface [91]. Although the TARF system still faces significant challenges, it provides an innovative solution for underwater-to-air communication and belongs to emerging technologies [92] with great potential for further improvement and integration with other methods.

2.13 Photoacoustic/thermoacoustic communication

Photoacoustic/Thermoacoustic communication (PAC/TAC) represents an advanced communication technology that utilizes the concept of using different mediums in air and water to achieve downward communication [93]. While TARF communication only supports upward communication, PAC/TAC allows for downward communication and serves as a complementary medium [94]. The theoretical basis of PAC/TAC is grounded in the photoacoustic or thermoacoustic effect, where the water surface absorbs pulsed electromagnetic energy, causing an increase in temperature and thus local thermal expansion, which produces an acoustic signal [95].

The On-Off Keying (OOK) modulation method is the primary technique used in PAC and TAC, with the binary “1” corresponding to the period during which an electromagnetic wave (light or microwaves) acts on the water surface, and “0” for the rest of the wave duration. Since this method can only send one bit at a time, the data rate is very low. Therefore, there is room for further exploration of more advanced modulation methods.

PAC and TAC are innovative technologies offering potential solutions to the problem of communication with underwater systems and can be integrated with existing communication technologies [96]. However, managing energy loss at the air-water interface remains a significant challenge that requires further research attention [97]. This issue is critical to optimizing the efficiency of communication systems as their success depends on the ability to transmit reliable signals between the aquatic and aerial environments. Improving

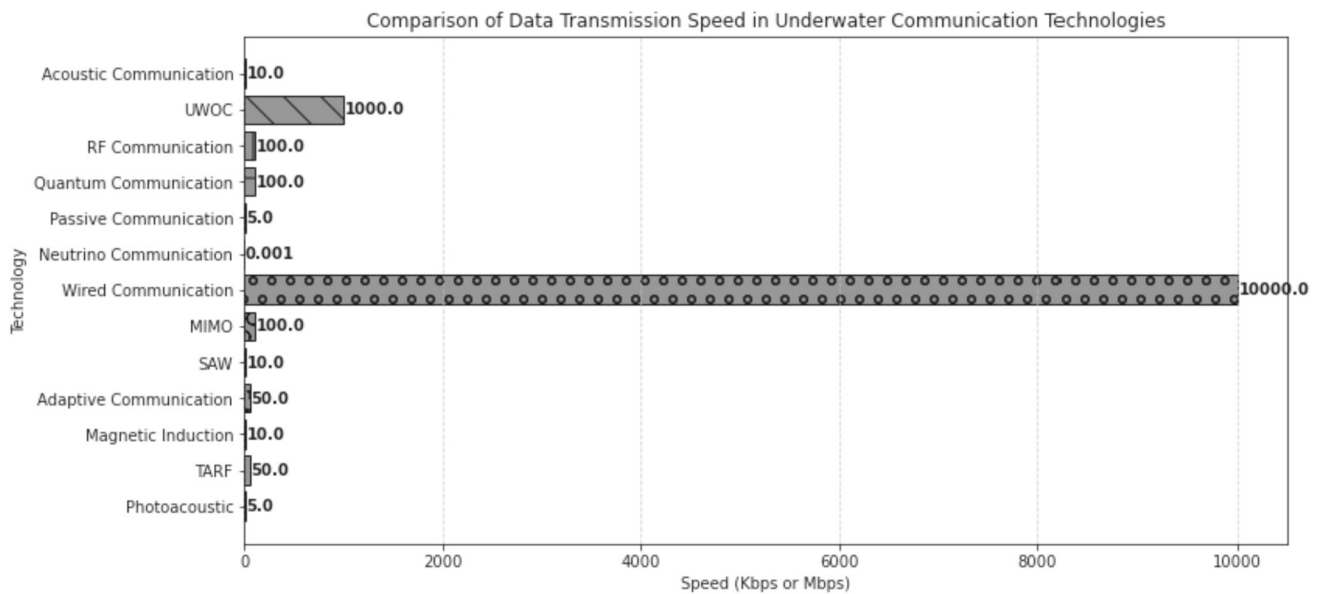


Fig. 1 Comparison of Data Transmission Speed in Underwater Communication Technologies. The chart highlights the significant variation in data transmission speeds (Kbps or Mbps) across different commu-

nication methods. Acoustic and RF communication offer lower speeds compared to fiber-optic and quantum-based systems

this technology could allow for more efficient and reliable communications, which is crucial for the safety and accuracy of underwater operations.

To further illustrate the differences among the existing underwater communication technologies presented in Table 1, the following figures provide a comparative analysis of key technical characteristics (Fig. 1).

These visual representations provide an intuitive way to compare different underwater communication technologies and complement the tabular data in Table 1. By analyzing the speed, range, and frequency characteristics, researchers and engineers can better understand which communication technology is most suitable for specific underwater applications (Fig. 2).

3 Innovative ideas and proposals

The advancement of underwater communication relies not only on refining existing technologies but also on the continuous development of innovative methods that address inherent challenges. Conventional techniques often struggle with issues such as high signal attenuation, limited transmission range, and environmental interferences, which hinder the effectiveness of underwater data exchange. To overcome these obstacles, researchers are exploring novel solutions that leverage emerging technologies, unconventional physical principles, and interdisciplinary approaches.

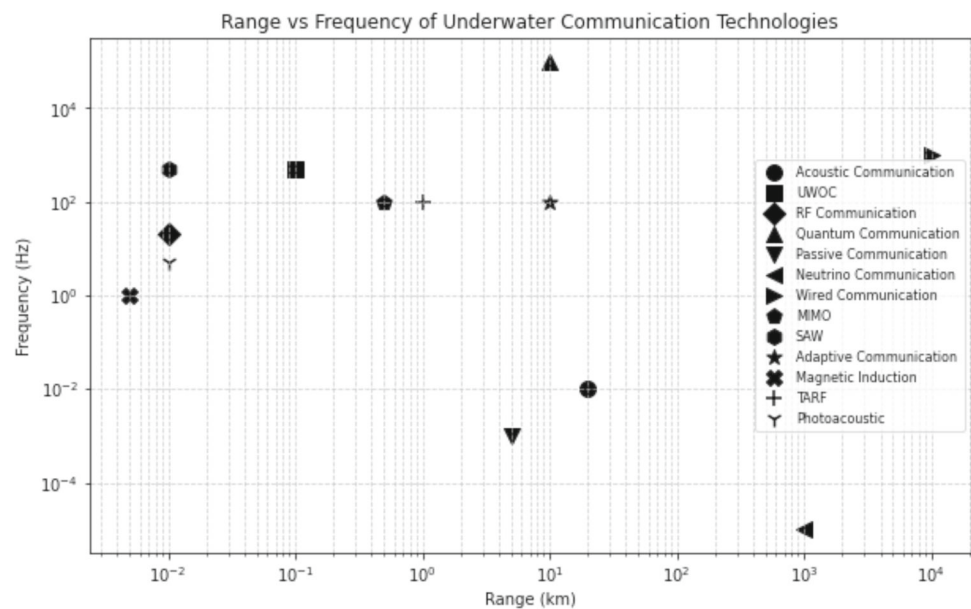
This section presents a selection of forward-thinking ideas and proposals that, while still in the research phase, have the potential to significantly enhance underwater communication capabilities. These innovations include the integration of artificial intelligence for adaptive signal processing, hybrid communication systems that combine acoustic, optical, and RF technologies, as well as biomimetic approaches inspired by marine life. Additionally, the application of advanced data encoding techniques, photonic communication, and novel physical transmission methods such as photoacoustic and metamaterial-based communication could lead to groundbreaking improvements in data transfer efficiency, security, and reliability.

A comparative analysis of these promising technologies is summarized in Table 2, offering insights into their key applications, advantages, and limitations. This systematic overview provides a structured evaluation of their potential impact and feasibility, highlighting the ongoing evolution of underwater communication and the research directions that may shape its future.

3.1 Hybrid communication systems

Hybrid communication systems represent a pioneering approach in underwater communication, as they combine various technologies, such as acoustic, optical, and radio waves, to create a more suitable and flexible underwater communication system [120]. This hybrid approach enables overcoming the limitations presented by each com-

Fig. 2 Range versus Frequency of Underwater Communication Technologies. This scatter plot visualizes the trade-offs between communication range and operating frequency, demonstrating how different technologies align with their intended underwater applications



munication method when used independently, providing a multi-faceted and multi-dimensional communication system.

The need for hybrid communication systems arises from the different challenges encountered in the underwater environment. Acoustic methods are ideal for long distances but are affected by noise and signal reflection. On the other hand, optical methods offer high transmission speeds and accuracy but are limited to short distances due to light absorption and scattering in water. Radio waves, while able to penetrate water to some extent, experience significant signal losses and are limited to very short-range communications.

Integrating these three technologies into a hybrid system provides an excellent solution for achieving great flexibility and robustness in various underwater conditions [121]. This ensures that communication can be maintained even in difficult conditions, where a single technology might fail. For instance, in cases where noise disrupts acoustic communication, the system can switch to optical or radio wave communication to maintain continuity. Moreover, the ability to combine data from various sources through a hybrid system can significantly improve the accuracy and reliability of transmitted information [122].

3.2 Advanced data encoding and compression algorithms

The advancement of underwater communication technologies has highlighted the need for more efficient data transmission methods [123, 124]. In this context, advanced data encoding and compression algorithms play a crucial role, offering the potential for more efficient data transmission with lower power consumption and increased performance

[125, 126]. These technologies enable the transmission of larger volumes of data through the aquatic environment with reduced information loss, paving the way for underwater communication and exploration [127].

The development and use of advanced data encoding and compression techniques aim to conserve available resources by reducing the time and energy required for data transmission [128]. Additionally, it allows for the transmission of more complex and information-rich data, such as videos [129, 130] or high-resolution images [131–133], which were previously difficult to transmit through the aquatic environment due to communication limitations.

3.3 Artificial intelligence for dynamic communication system adjustment

The use of artificial intelligence (AI) in underwater communication systems can lead to more flexible, automated, and efficient solutions. AI can play a critical role in dynamically adjusting underwater communication systems, enabling automatic adaptation to environmental conditions and communication requirements [134].

Specifically, an automated communication system can use AI algorithms to analyze water conditions, such as turbidity and conductivity, and dynamically adjust the communication method used, switching from acoustic to optical communication or even to radio frequencies, depending on which will offer the best performance under those specific conditions. This adaptive capability allows successful data transmission with the highest possible efficiency and minimal information loss [135].

Furthermore, AI can be applied for real-time error detection and correction, optimization of data compression algo-

Table 2 Comparative table of future expansions and innovations in underwater communication

Communication type	Usage	Advantages	Limitations	Technical characteristics
Hybrid Communication Systems [120–122]	Combination of acoustic, optical, RF	Flexibility, adaptability to environments	Complexity, high cost	Speed: 10 kbps–1 Gbps, Range: 1 m–20 km, Frequency: Variable
Advanced Data Encoding [123–133]	Enhanced data transfer	Reduced bandwidth consumption, improved efficiency	Dependent on communication technology	Varies by underlying technology
AI for Dynamic Tuning/Adaptation [134–136]	Automated optimization	Real-time adaptation	High computational demands	Speed and range depend on communication system
Internet of Underwater Things (IoUT) [137, 138]	Device interconnection	Connectivity, monitoring	Infrastructure dependency	Speed: 10 kbps–100 Mbps, Range: 10 m–10 km, Frequency: Device-dependent
Biological Carriers [17, 139]	Using natural phenomena	Low energy consumption	Limited speed	Speed: 1–5 kbps, Range: 1 km, Frequency: Natural biology
Photonic Communication [140, 141]	Underwater networks	High speed	Limited range	Speed: 10 Mbps–1 Gbps, Range: 10–100 m, Frequency: 400–800 THz
High-Frequency Ultrasonic Waves [142, 143]	Underwater networks	Higher speed than acoustic communication	Shorter range	Speed: 10–50 kbps, Range: 1–5 km, Frequency: 100 kHz–1 MHz
Acoustic Metamaterial Communication [144, 145]	Noise reduction	High efficiency	Experimental stage	Speed: 5–20 kbps, Range: up to 10 km, Frequency: Variable
Photoacoustic Signal Communication [146, 147]	Short distances	Low energy consumption	Limited range	Speed: 1–10 kbps, Range: 1–10 m, Frequency: Variable
Networked AUV Swarms with Swarm Technology [146–171]	Coordinated swarm operations	Self-organization, resilience	Implementation complexity	Speed: 10 kbps–100 Mbps, Range: 1–20 km, Frequency: Technology-dependent
Biomimetic Methods [18, 19, 172–190]	Nature-inspired underwater communication	Low energy, stealth operation, silent	Low speed, short range	Speed: 1–100 bps, Range: 1–20 m, Frequency: Method-dependent

Table provides a comparative summary of the most promising future directions in underwater communication. It outlines innovative methodologies that have the potential to revolutionize the field, ranging from hybrid systems to biomimetic methods inspired by marine organisms. Each approach is assessed based on its primary applications, benefits, and inherent challenges, offering insights into their feasibility for real-world implementation

rhythms to reduce energy requirements for transmission, and increase the overall efficiency of the system [136]. Advanced data analysis and processing using AI technologies can also identify patterns and trends not easily perceptible to humans, thus improving communication performance and data transfer accuracy.

3.4 Underwater communication via quantum entanglement

Quantum entanglement, a phenomenon where two or more quantum particles become so closely connected that the state of one directly affects the state of the other, even over long distances, offers the foundation for underwater communication through entanglement [191]. The practical application

of this in the underwater environment could rely on specially designed quantum sensors and emitters that create and emit entangled quantum particles, enabling secure information transfer from the seabed to the surface or vice versa [192–195].

This technology could allow the implementation of an integrated and secure communication system for various purposes, including scientific research, environmental monitoring, and military surveillance. Through quantum entanglement, information could be transmitted at speeds surpassing current methods, while ensuring absolute data security due to the impossibility of interception or breaching quantum encryption without causing detectable disturbance in the particle states.

3.5 Networking of underwater devices with IoT (Internet of Underwater Things)

Integrating underwater systems into the Internet of Things (IoT) enables the creation of a connected network (Internet of Underwater Things - IoUT) that facilitates real-time data collection and exchange among devices, enhancing monitoring, management, and security of underwater areas [196–198].

Integrating underwater systems within the broader IoT community opens new avenues for the development of IoUT [199, 200]. By networking underwater devices, a connected ecosystem allows seamless data collection and exchange in real-time [137, 201]. For example, a network of underwater sensors can measure various water parameters, such as temperature, pressure, and chemical composition, sending the information to a central database or even a smartphone on the surface for immediate analysis and operational decision-making [202, 203].

3.6 Underwater acoustic transformation communication technology

Underwater Acoustic Transformation Communication technology is based on the idea of creating and emitting acoustic signals that can dynamically change their properties—such as frequency, amplitude, or duration—based on the communication content. This enables data transfer in a way that is very difficult for unauthorized users to detect or intercept. For example, a communication system using this technology can send acoustic signals with specially encoded patterns that change according to the nature of the transmitted information, significantly increasing transmission security [204, 205].

Additionally, the use of acoustic transformation technology can improve the efficiency of underwater communication by addressing signal loss and interference challenges common in the aquatic environment [138]. The ability to adjust signal characteristics allows better management of the acoustic

properties of water and adaptation to different conditions prevalent in the underwater environment [206, 207].

3.7 Underwater communication through biological carriers

The concept of underwater communication through biological carriers involves the possibility of using biological organisms, such as microbes or modified cells, as natural data carriers. Using microbes or genetically modified cells, it is possible to create natural “carriers” that can be programmed to respond to specific stimuli by producing particular patterns or signals [208]. Specifically, a particular sequence of genes could be activated in the presence of certain chemicals, causing the microbes to emit light signals or other detectable patterns as a means of communication [209].

This approach has the advantage of high energy efficiency and sustainability, as it utilizes biological organisms that can reproduce and sustain themselves naturally in the aquatic environment. Furthermore, using such biological systems could provide a secure form of communication, as the detection and decoding of patterns or signals by unauthorized parties would require specialized knowledge and technology. These technologies could be applied to applications involving water quality monitoring, pollution source identification, or even scientific research to study underwater ecosystems [17]. Additionally, the biological nature of this communication could contribute to the development of new methods for interacting with marine animals, supporting efforts for the protection and sustainable management of marine resources.

3.8 Utilizing underwater photons for communication

The use of underwater photons for communication represents one of the most advanced and intriguing prospects in the field of underwater communications [210]. This method leverages the property of photons to retain their energy as they propagate through water, allowing efficient information transmission with minimal losses [211]. Specially designed light sources, such as lasers or LEDs [212], can be used to create coded photonic signals that can be detected and decrypted by receivers over considerable distances, thereby enabling high-quality offshore communication [139, 213, 214].

To implement this idea, advanced coding and decryption systems could be developed, harnessing photon technology to achieve secure and efficient data transmission [215]. Furthermore, the integration of technologies such as Quantum Key Distribution (QKD) could enhance communication security, making it ideal for applications requiring maximum security, such as military or research missions.

3.9 Underwater communication via high-frequency sound waves

Key elements of underwater communication via high-frequency sound waves include the development and application of specialized technologies that allow rapid data transmission through the aquatic environment. High-frequency sound waves are particularly preferred because they provide greater clarity and less data loss compared to traditional methods [216]. This makes the technology ideal for critical applications, such as communication between Autonomous Underwater Vehicles (AUVs), underwater robotics, and environmental data monitoring.

A crucial aspect of these systems' development is the integration of modern data compression algorithms, enabling the more efficient transmission of large volumes of data at lower energy costs [217]. Additionally, advanced encoding techniques allow the selection of appropriate frequencies for transmission depending on the underwater environment conditions, such as acoustic clarity and the presence of obstacles [140, 218].

3.10 Underwater contextual wave communication

Underwater contextual wave communication highlights an innovative approach to data transmission, where the natural wave patterns of water exploit the movement and properties of the aquatic environment for information transfer [219, 220]. Utilizing technologies for recording and analyzing wave patterns, this method aims to develop a data integration system capable of converting natural wave variations into information [221].

Research focuses on creating encoded "signatures" or patterns in waves, which can be interpreted and decoded as data signals. This allows communication over significantly larger distances beneath the sea surface, reducing the need for high energy power [141].

3.11 Underwater communication through discrete thermal signals

Communication through discrete thermal signals explores a groundbreaking approach to underwater data transmission by using temperature changes in water as a means of communication [222–225]. This method is developed using specially designed systems that produce and detect local thermal changes, creating a coded "temperature alphabet." This alphabet comprises various thermal change patterns emitted into the aquatic environment, which can efficiently convey information through differentiated thermal signals [142].

3.12 Integration of communication systems with autonomous underwater vehicles (AUVs)

Based on the necessity for reliable and efficient communication in underwater environments, incorporating specialized communication systems into Autonomous Underwater Vehicles (AUVs) offers significant potential for enhancing their capabilities. Developing such systems enables full utilization of AUVs as mobile communication nodes, making them capable of efficiently collecting and exchanging data in real-time [226–229].

3.13 Using underwater photoacoustic signals for communication

Exploration of the potential to combine light and sound to create photoacoustic signals that can be transmitted and detected underwater presents an alternative method for data transmission through water [143]. This technique may overcome the limited ranges of visibility and audibility prevailing underwater, providing a more efficient alternative for complex communications [230]. This methodology relies on creating photoacoustic waves by stimulating water with high-intensity lasers, which then propagate as acoustic signals [14, 144, 231, 232].

3.14 Interactive underwater communication via smart materials

Developing smart materials [233–235] and sensors [12] that can alter their physical properties (e.g., color, transparency, light emission) [145] in response to electronic stimuli [236] or other external signals [237] offers promising opportunities for creating underwater communication networks [238] that can adapt and respond to varying conditions and requirements.

One emerging approach in this field is the integration of acoustic meta-materials, which are engineered to manipulate sound waves in ways that enhance underwater communication. Recent advancements in multi-oriented acoustic meta-materials have demonstrated their potential for omnidirectional sound wave absorption, improving signal clarity and reducing interference in underwater environments [239]. By utilizing these advanced materials, underwater communication systems can achieve greater efficiency in signal transmission, particularly in noisy or highly dynamic marine conditions. These smart materials, combined with bio-inspired technologies, could lead to innovative solutions for adaptive underwater networks, facilitating real-time modulation of communication properties in response to environmental stimuli.

3.15 Swarm of AUVs for enhanced communication

The challenges of the underwater environment [148], such as high signal attenuation, varying conditions, and difficulty in achieving reliable communication over long distances, underscore the need for innovative solutions [146, 149]. Autonomous Underwater Vehicles (AUVs) operating in swarm formations present a promising approach to overcoming these limitations by leveraging coordinated movement and decentralized communication strategies. Applying technologies such as ultrasonic communication, laser-based optical communication, and swarm technology [150] enables AUVs to form complex communication networks [151] that self-organize and adapt to changing conditions. Advances in underwater robotics and multi-agent control systems play a crucial role in optimizing swarm-based communication, allowing for enhanced resilience and adaptability in dynamic underwater environments [240].

Furthermore, the implementation of advanced control strategies, such as finite-time observer-based tracking and adaptive sliding mode approaches, has been shown to improve the stability and efficiency of heterogeneous underwater vehicle coordination [241]. These techniques enable precise maneuvering and robust communication among AUVs, ensuring reliable data exchange even in challenging underwater conditions. By integrating these technologies, AUV swarms can efficiently exchange information, enhance security [152], and improve the reliability of underwater missions [147, 153, 154]. Future research aims to refine these strategies through AI-driven optimization and self-learning algorithms, further enhancing the autonomy and adaptability of underwater robotic networks.

3.15.1 Underwater communication via ultrasonic pulses

Development of a technology using ultrasonic pulses for data transmission between AUVs offers a reliable communication method over long underwater distances, with the ability to penetrate complex or challenging underwater environments [155].

3.15.2 Interconnected AUVs with optical communication

Utilizing optical communication, such as lasers or LEDs, to create a communication network among AUVs can provide high transmission speeds and increased security, as optical links are harder to detect or intercept by third parties [156].

3.15.3 Dynamic communication networks among AUVs via swarm technology

Leveraging swarm technology to create dynamic communication networks among AUVs allows these networks to

adapt and change dynamically, optimizing communication coverage and efficiency based on current missions and environmental conditions [157]. This approach can facilitate real-time data collection and sharing from various AUVs, enhancing collaboration and the effectiveness of collective missions.

3.15.4 Enhanced communication through advanced environmental modeling and recognition

Developing technologies that enable AUVs to recognize and model the underwater environment in real time through artificial intelligence and machine learning can significantly enhance data transmission efficiency and reliability [158]. Such advancements would allow AUVs to adjust their communication strategies based on detected conditions, improving overall performance [159].

3.15.5 Energy-based underwater communication networks

This approach entails developing systems that enable AUVs to create energy-efficient communication networks by leveraging energy detection technologies [160] or generation [161]. This may involve using environmental sensors to detect energy fields or differences related to natural oceanic processes, such as temperature variations, to optimize communication and conserve energy [162–164].

3.15.6 Exploiting natural acoustic resonance for AUV communication

By harnessing the natural acoustic resonance abilities of underwater structures and environments, this technology could develop methods for encoding and decoding information through acoustic resonance [165, 166]. This offers an efficient communication method between AUVs navigating specific underwater areas.

3.15.7 Hybrid communication networks combining acoustic and optical technologies

This communication system exploits both acoustic [167, 168] and optical technologies to establish a broadband and reliable network for AUVs. The hybrid approach could allow efficient communication in various underwater environments by adjusting the communication technology choice based on conditions such as visibility and noise levels [169].

3.15.8 Self-organizing networks (SONs) for underwater AUVs

Applying Self-Organizing Networks (SONs) for underwater AUVs represents an innovative approach in the communica-

tion and management of underwater systems. Through this technology, AUVs can form networks that dynamically self-organize and adapt to changes in the underwater environment and mission requirements, improving efficiency and reliability without constant external intervention [170].

SONs allow for autonomous network management, including device auto-discovery, dynamic network restructuring to handle disruptions or environmental changes, and automated optimization of communication for maximizing energy efficiency and coverage [171]. This adaptability is expected to enhance the effectiveness of underwater missions, offering flexible and reliable solutions for exploration, monitoring, and management.

3.16 Biomimetic methods

This section explores potential research extensions in underwater communication based on biomimicry principles. This approach draws inspiration from nature and natural processes [172], aiming to replicate or mimic biological communication methods observed in marine organisms. From utilizing light-coded underwater bubbles to leveraging bioelectric fields, these methods reflect the complexity and diversity of natural communication mechanisms.

Each of the following subsections delves into a specific biomimetic technique, revealing its full potential for revolutionary applications in underwater communication.

3.16.1 Light-coded underwater bubble communication

Idea: Utilize the bubble phenomenon in water, combined with light technology that emits or changes light during bubble formation, to convey information [173].

This idea presents a novel underwater communication concept, leveraging the natural ability of AUVs to generate bubbles. The core idea involves using light signals, such as LEDs, that change colors or produce specific light patterns during bubble formation, thereby encoding different information into each bubble [174].

The proposed communication method operates by emitting bubbles from an AUV, each illuminated by an embedded light source capable of changing colors or light patterns depending on the information to be conveyed. Another AUV or an underwater detection station can then “read” these bubbles, identifying specific light patterns as distinct information. This method offers advantages such as simplicity, flexibility, and security, while addressing challenges like visibility and accuracy. The application of this technology would require significant advancements in lighting technology and detection sensor development [175] to successfully read the encoded information. Additionally, algorithm and communication protocol development [176, 177] is necessary to

effectively manage the transmission and recognition of different signals.

3.16.2 Bioluminescent signal communication

By utilizing the natural ability of certain marine organisms to emit light (bioluminescence) [178], a communication system could be developed that leverages this ability for underwater information transfer. This approach involves developing biomimetic luminous sensors that imitate the bioluminescence of marine organisms, such as phosphorescent plankton, capable of emitting light signals at various frequencies or colors. These signals can then be used for encoding and transmitting information [179], following a predefined system that translates different frequencies or colors into specific information [180, 181].

This communication system offers multiple benefits, primarily ecological compatibility, as it utilizes a natural, biological process for communication, reducing environmental impact [182]. Additionally, this method allows for signal detection and decoding by AUV sensors, enhancing reliability and efficiency [183].

3.16.3 Underwater acoustic mimicry communication

By leveraging the ability of certain marine animals, such as dolphins, whales, and certain fish species, to produce and use complex sounds for communication, we could develop a system that mimics these natural acoustic patterns [184]. Advanced underwater acoustic transformations provide new ways to create and emit sound signals that can dynamically change their properties, such as frequency, amplitude, and duration, based on communication needs [185].

This approach aims to develop artificial acoustic mimics that produce sound patterns similar to those of marine animals [18], embedded in AUVs to encode and transmit information through varied frequencies, durations, and sound patterns [186]. This communication method offers a new dimension in underwater interactive communication, allowing AUVs to exchange data and commands through a sophisticated, yet naturally inspired, technique.

3.16.4 Underwater bioelectric field communication

Utilizing bioelectric communication, mimicking how some underwater creatures, such as electric eels, produce and use electric fields to orient themselves, communicate, or detect prey in their aquatic environment [187].

This method may involve the development of technology [19] that enables AUVs to generate and emit electrical signals, as well as special sensors to detect and interpret these signals [188]. These sensors and associated processing algorithms allow recognition and decoding of informa-

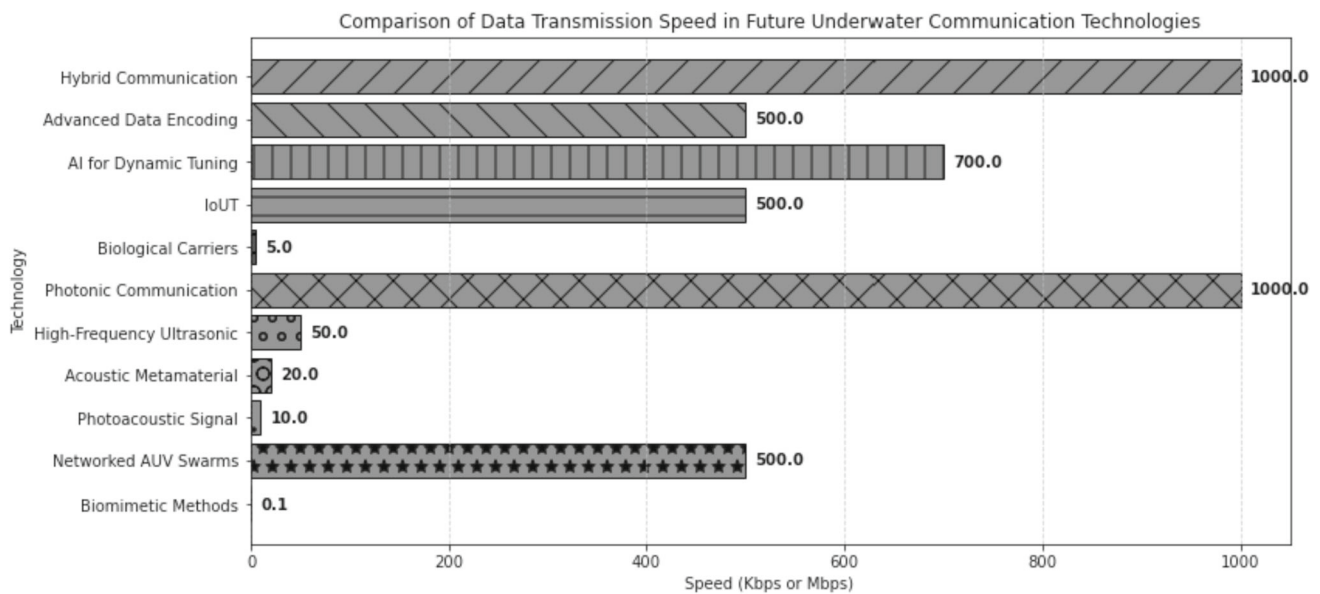
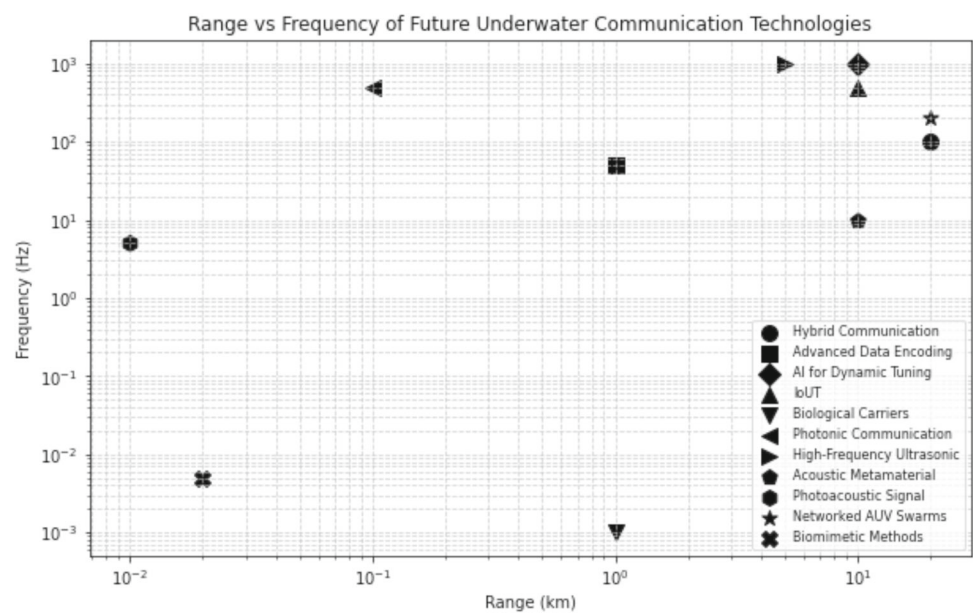


Fig. 3 Comparison of Data Transmission Speed in Future Underwater Communication Innovations. The chart showcases the expected data transmission capabilities of emerging technologies, highlighting potential advancements over traditional methods

Fig. 4 Range versus Frequency in Future Underwater Communication Technologies. This scatter plot visualizes how innovative technologies balance range and operating frequency, offering insights into their feasibility and applications



tion transmitted through bioelectric signals, offering a new dimension in underwater communicative interaction [189, 190].

To provide a clearer understanding of the future expansions and innovations outlined in Table 2, the following figures illustrate key technological characteristics and their comparative advantages (Fig. 3).

These visualizations supplement Table 2 by illustrating the comparative characteristics of emerging technologies. By analyzing the expected transmission speeds, ranges, and frequencies, researchers can gain a better understanding of

which technologies hold the most promise for the future of underwater communication (Fig. 4).

4 Research challenges and open issues

Despite the rapid advancements in underwater communication technologies, several fundamental challenges remain unresolved, hindering their widespread adoption and efficiency. The unique characteristics of the underwater environment introduce significant constraints, including signal attenuation, multipath interference, and energy limitations, which

necessitate continuous research and innovation. Addressing these challenges is critical for enabling reliable, scalable, and secure communication networks that support applications in marine exploration, environmental monitoring, defense, and deep-sea industrial operations.

This section discusses the key obstacles that researchers must overcome to improve underwater communication systems. It explores the impact of environmental conditions on signal propagation, the constraints imposed by energy efficiency and power availability, and the increasing need for secure data transmission. Additionally, the scalability and interoperability of underwater networks pose significant concerns, particularly as the concept of the Internet of Underwater Things (IoUT) gains momentum. Furthermore, emerging technologies such as quantum communication, biomimetic strategies, and AI-driven adaptive systems present new opportunities to revolutionize underwater communication. By examining these open research issues, this section highlights the gaps that need to be addressed to advance the field and ensure the development of robust, high-performance underwater communication systems. These challenges not only impact current implementations but also shape the future direction of research, influencing the design of next-generation underwater networks that will push the boundaries of technological innovation.

4.1 Physical limitations and environmental factors

One of the most significant challenges in underwater communication is the impact of environmental factors on signal propagation. Acoustic communication, despite being the most widely used method, suffers from high attenuation, multipath interference, and long propagation delays due to the density and variability of the underwater medium. Optical and RF-based communication systems offer high data rates but are severely limited by water turbidity, absorption, and scattering. Future research must focus on hybrid systems that intelligently switch between communication modes based on real-time environmental conditions to optimize performance.

Moreover, the complex and dynamic nature of underwater environments poses challenges related to noise interference from natural and anthropogenic sources. For instance, marine life, geological activities, and human-made structures contribute to background noise that can degrade signal quality. Advanced signal processing techniques, such as machine learning-driven noise reduction and adaptive filtering, can play a crucial role in mitigating these effects and enhancing underwater communication efficiency.

4.2 Energy efficiency and power constraints

The power limitations of underwater devices present another major challenge. Unlike terrestrial communication systems,

where energy sources are readily available, underwater communication relies on battery-powered systems with restricted energy reserves. Long-duration operations, such as autonomous underwater vehicle (AUV) missions and deep-sea sensor networks, demand energy-efficient communication protocols to maximize operational longevity.

Research is needed to develop energy-efficient modulation schemes, power-aware networking strategies, and energy harvesting techniques, such as ocean thermal energy conversion (OTEC) and bio-inspired power generation. The integration of self-sustaining power sources with underwater communication networks could significantly enhance operational capabilities and reduce the need for frequent maintenance and battery replacements.

4.3 Security and data integrity concerns

Underwater communication systems are increasingly being deployed for critical applications, including military defense, environmental monitoring, and deep-sea exploration. Ensuring secure and reliable data transmission in such scenarios is paramount. However, the underwater domain presents unique cybersecurity challenges due to the lack of standardized encryption protocols and the vulnerability of acoustic signals to interception and jamming attacks.

Future research must focus on developing robust encryption mechanisms tailored for underwater environments, leveraging quantum cryptography, blockchain-based security frameworks, and AI-driven anomaly detection to prevent unauthorized access and data breaches. Additionally, secure authentication methods and intrusion detection systems must be integrated into underwater networks to safeguard sensitive information.

4.4 Scalability and interoperability of networks

As the concept of the Internet of Underwater Things (IoUT) gains traction, ensuring seamless interoperability and scalability of underwater communication networks becomes increasingly critical. Current underwater networks operate in isolated deployments, often using proprietary protocols that limit cross-compatibility. Standardization efforts are required to develop unified communication frameworks that allow different underwater devices, AUVs, and sensor nodes to interconnect efficiently.

Additionally, scalable architectures that support dynamic network expansion and real-time adaptation to varying communication demands must be explored. The use of AI-driven network optimization, software-defined networking (SDN), and cognitive radio techniques can facilitate adaptive communication and improve resource allocation in underwater networks.

4.5 Future directions and emerging technologies

The future of underwater communication lies in leveraging emerging technologies to overcome existing limitations and unlock new capabilities. Quantum communication, for example, holds the potential to revolutionize secure data transmission by utilizing entanglement-based communication principles. While still in the experimental stage, quantum techniques could offer unprecedented security and reliability for underwater applications.

Moreover, the integration of biomimetic communication strategies inspired by marine organisms, such as bioluminescent signaling and electric field-based messaging, could provide low-energy alternatives for underwater data exchange. Research into advanced AI-driven communication protocols, self-healing networks, and deep-sea optical relay systems will be instrumental in shaping the next generation of underwater communication technologies.

In conclusion, while significant progress has been made in underwater communication, numerous research challenges and open issues must be addressed to realize fully operational and efficient underwater networks. By tackling environmental constraints, power limitations, security vulnerabilities, and network scalability issues, researchers can pave the way for transformative advancements in the field, ensuring robust and resilient underwater communication systems for future applications.

5 Discussion

Underwater communication technologies are undergoing a significant transformation, driven by the demand for more efficient, secure, and adaptable solutions. The evolution of acoustic, optical, and radio frequency-based methods, alongside novel approaches such as quantum and biomimetic communication, has paved the way for innovative systems capable of addressing long-standing limitations. These advancements are not only expanding the frontiers of scientific research but also supporting critical applications in maritime security, deep-sea exploration, environmental monitoring, and industrial operations. However, despite the remarkable progress, numerous challenges remain, necessitating interdisciplinary research and engineering breakthroughs.

One of the most promising developments in this domain is the increasing integration of artificial intelligence (AI) and machine learning into underwater networks. These technologies offer adaptive communication strategies that enhance real-time data processing, predictive maintenance, and self-optimizing transmission protocols. AI-driven approaches, such as reinforcement learning for dynamic spectrum allocation and intelligent signal processing, improve both the

efficiency and reliability of underwater networks. Additionally, AI plays a crucial role in the coordination of autonomous underwater vehicles (AUVs), enabling real-time decision-making and adaptive navigation in complex underwater environments. The convergence of AI with underwater communication is expected to redefine how data is transmitted and processed in subsea operations, making autonomous systems more resilient and responsive to environmental changes.

Another pivotal trend is the shift toward hybrid communication systems, which combine multiple modalities to achieve seamless data transmission under varying conditions. Hybrid systems leverage the long-range capabilities of acoustic signals, the high-speed transmission of optical links, and the unique advantages of electromagnetic waves, optimizing performance based on environmental parameters. The implementation of intelligent switching mechanisms between different communication modes ensures that data transmission remains robust even in the presence of interference, signal attenuation, or dynamic topographical constraints. The development of such multi-layered networks will be instrumental in large-scale underwater infrastructure, where reliability and redundancy are crucial.

Furthermore, biomimetic communication methods inspired by marine life introduce novel solutions for sustainable and interference-free data exchange. Organisms such as electric fish, bioluminescent species, and cetaceans have evolved highly efficient natural communication strategies that can be adapted to human-engineered systems. By mimicking these biological processes, researchers are exploring energy-efficient communication methods that minimize disruptions to marine ecosystems. These innovations align with the broader goal of eco-friendly underwater technologies, ensuring that the expansion of underwater communication does not come at the expense of biodiversity or environmental stability.

Despite these advancements, key bottlenecks persist, particularly in terms of energy consumption, signal attenuation, and scalability. The high power requirements of optical and RF communication systems, the limited range of high-frequency signals, and the impact of environmental variability on transmission quality pose significant barriers to widespread deployment. Additionally, the lack of standardized protocols for Internet of Underwater Things (IoUT) remains a fundamental challenge, hindering seamless interoperability between underwater devices from different manufacturers and research institutions. Addressing these limitations will require collaborative efforts from the fields of ocean engineering, telecommunications, artificial intelligence, and materials science.

Looking ahead, the future of underwater communication is poised to benefit from advancements in quantum technologies, photonics, and self-organizing networks. Quantum entanglement-based communication could revolutionize data

security in underwater environments, eliminating vulnerabilities associated with traditional encryption methods. Similarly, advancements in photonic communication may lead to faster, more energy-efficient systems that overcome the limitations of conventional optical links. The deployment of self-organizing underwater networks, inspired by terrestrial ad hoc systems, could further enhance the reliability and adaptability of underwater communication infrastructures.

In conclusion, the ongoing development of underwater communication technologies is shaping the future of ocean exploration, defense, resource management, and climate monitoring. While significant progress has been made, continued research is essential to overcome existing constraints and unlock the full potential of these systems. By integrating AI, hybrid networks, biomimetic approaches, and cutting-edge quantum technologies, the field is set to redefine our ability to interact with and understand the underwater world. The future lies in interdisciplinary innovation, where collaboration across multiple domains will drive the next generation of breakthroughs in underwater communication.

Author Contributions All authors contributed to the study conception and design. All authors read and approved the final manuscript.

Funding Open access funding provided by HEAL-Link Greece. No funding was received for conducting this study.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors have no Conflict of interest to declare that are relevant to the content of this article.

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